

Subject Description Form

Subject Code	EE3013																					
Subject Title	Transportation Data Analytics																					
Credit Value	3																					
Level	3																					
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite of EE3013: EE2029 / EEE2003, CSE30390 Pre-requisite of EE3013B: EE2029B, CSE30390																					
Objectives	1. To introduce various types of transportation data, and ways to use the data to assess, analyze, and assist the modeling of transportation systems. 2. To equip the students with modeling and analysis techniques for transportation data. 3. To enable the students to understand problems and issues in real-world transportation data and methods to deal with them. 4. To prepare the students for tackling real-world transportation problems using data, with a combination of deep understanding of data issues and solid analytical skills.																					
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Demonstrate theoretical knowledge of transportation data analytics b. Apply appropriate data analytics methods and tools to various types of transportation data and interpret the results c. Understand problems and issues in real-world data and ways to tackle those problems and issues																					
Subject Synopsis/ Indicative Syllabus	1. Probability, statistics, and regression for transportation data 2. Choice model, modeling travel behavior using travel survey data 3. Data collection, analysis, and visualization using Python 4. Machine learning models for transportation data																					
Teaching/Learning Methodology	Delivery of the subject is mainly through formal lectures and complemented by tutorials. Exercises (in-class or take-home), assignments, and projects provide students hands-on experience in data modelling, estimation, and analysis of practical transportation problems, while report-writing enables students to practise writing skill. <table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th rowspan="2">Teaching/Learning Methodology</th><th colspan="3">Outcomes</th></tr> <tr> <th>a</th><th>b</th><th>c</th></tr> </thead> <tbody> <tr> <td>Lectures</td><td style="text-align: center;">✓</td><td style="text-align: center;">✓</td><td style="text-align: center;">✓</td></tr> <tr> <td>Tutorials</td><td style="text-align: center;">✓</td><td style="text-align: center;">✓</td><td style="text-align: center;">✓</td></tr> <tr> <td>Exercises, assignments and projects</td><td style="text-align: center;">✓</td><td style="text-align: center;">✓</td><td style="text-align: center;">✓</td></tr> </tbody> </table>			Teaching/Learning Methodology	Outcomes			a	b	c	Lectures	✓	✓	✓	Tutorials	✓	✓	✓	Exercises, assignments and projects	✓	✓	✓
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Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed		
			a	b	c
	1. In-class exercises and assignments	30%	✓	✓	✓
	2. Projects	40%	✓	✓	✓
	3. Midterm exam	30%	✓	✓	✓
	Total	100%			
The midterm examination allows assessment on outcomes covering concepts, principles, techniques and application for the first half of teaching focusing on theories. Exercises, assignments and projects enable students to explore and apply analytical and tool-based data modelling techniques to evaluate transportation systems’ characteristics and performance. Projects are mainly arranged in the second half of the semester, reflecting the steering of focus from theories to practice. Report-writing enables students to interpret the data analysis results, link them to practical issues in transportation systems and find solutions.					
Student Study Effort Expected	Class contact:				
	▪ Lecture/Tutorial	39 Hrs.			
	Other student study effort:				
	▪ Exercises, assignments and projects	35 Hrs.			
	▪ Self-study	33 Hrs.			
	Total student study effort		107 Hrs.		
Reading List and References	1. Richard J. Larsen and Morris L. Marx, An Introduction to Mathematical Statistics and Its Applications, 5th Edition, Prentice Hall, 2012.				
	2. Robert S. Pindick and Daniel L. Rubinfeld, Econometric Models and Economic Forecasts, 4th Edition, Irwin/McGraw-Hill, 1998.				
	3. Frank S. Koppelman and Chandra Bhat, A Self Instructing Course in Mode Choice Modeling: Multinomial and Nested Logit Models, 2006. https://www.ce.utexas.edu/prof/bhat/courses/lm_draft_060131final-060630.pdf				
	4. Jeremy Watt, Reza Borhani and Aggelos K. Katsaggelos, Machine Learning Refined: Foundations, Algorithms, and Applications, Cambridge University Press, 2016.				
	5. Marco Gori, Machine Learning: A Constraint-Based Approach, Morgan Kaufmann, 2017.				